

RENAL SYSTEM

NCLEX STUDY GUIDE

Assessment • Laboratory patterns • Dialysis • Safety

1. Testicular and Scrotal Disorders

Condition	Key findings / management
Testicular torsion	Emergency: sudden excruciating pain and absent cremasteric reflex; urgent surgery.
Cryptorchidism	Undescended testis, often related to prematurity; many descend by 6 months. Persistent cases need repair to reduce infertility/cancer risk.
Testicular cancer	Age 15-30; painless slow-growing lump, possible gynecomastia, increased AFP/hCG; history of cryptorchidism.
Hydrocele	Fluid in scrotal sac, common in newborns; swelling transilluminates, often no treatment.
Epididymitis	Fever/chills, scrotal or groin pain, pyuria; treat infection and sexual partner when STI-related, fluids and cold support.

2. Kidney Stones and Incontinence

Stone type	Diet guidance
Calcium oxalate	Limit high-oxalate foods such as cocoa, tea, cola, beans, nuts, rhubarb and some greens; cranberry juice may be used if appropriate.
Uric acid	Limit purine-rich organ meats, sardines, herring, salmon, bacon, pork, beef and shellfish; avoid cranberry juice when urine acidification is undesirable.

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- Renal colic: severe pain radiating to groin/genitals, nausea/vomiting and hematuria.
- Hydrate about 2.5-3 L/day when not contraindicated, provide analgesia, ambulate and strain urine.
- After lithotripsy, flank bruising may occur; hold anticoagulants/NSAIDs beforehand as ordered.

Type	Typical pattern / intervention
Stress	Leak with pressure/cough; Kegel exercises and prescribed therapy.
Urge	Sudden urgency/overactive bladder; bladder training, double voiding, reduce caffeine/alcohol; oxybutynin may be prescribed.
Overflow/reflex	Neurogenic bladder; Crede/Valsalva as instructed, bethanechol (never IV/IM), intermittent clean catheterization.
Functional	Physical/cognitive barrier; scheduled toileting and accessible clothing/environment.

3. Glomerular Disorders

Glomerulonephritis	Nephrotic syndrome
Often 2 weeks after streptococcal infection; periorbital edema, cola-colored urine, proteinuria and hypertension. Priority: altered LOC may indicate hypertensive encephalopathy. Restrict sodium, potassium and fluids; treat cause and use seizure precautions.	Massive proteinuria (>3 g/day), generalized edema/ascites, low urine output, frothy urine and hyperlipidemia. Maintain normal protein intake, restrict sodium/fluids as ordered, use steroids and monitor infection.

4. Urinary Tract Infection and Prostate

Disorder	Key points
Cystitis	Dysuria/burning; fluids, phenazopyridine and antibiotic as ordered. Nitrofurantoin may discolor urine; stop/report pulmonary symptoms.
Pyelonephritis	Flank pain and CVA tenderness, fever/chills, nausea, dysuria and cloudy/foul urine; elderly may present with delirium. Culture confirms; treat with antibiotics and fluids.
Prostatitis	Fever/chills, dysuria, low-back pain, discharge and warm/boggy prostate; fluids, sitz bath, antibiotics and stool softener; avoid prostate massage.
BPH	Nocturia, hesitancy, weak stream, dribbling and retention. Finasteride/dutasteride or alpha blockers; monitor orthostatic hypotension.

TURP priorities

- CBI triple-lumen catheter; pink drainage and small clots may be expected initially. Bright-red blood or large clots require notification.
- Bladder spasm may indicate clots/obstruction; irrigate only as prescribed.
- TURP syndrome: fluid overload and dilutional hyponatremia causing confusion, seizure or cerebral edema.

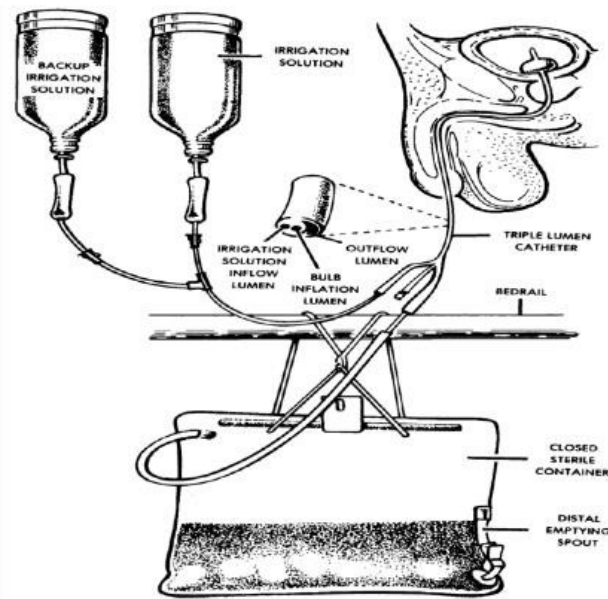


Figure 2-4. Continuous bladder irrigation.

Triple-lumen catheter used for continuous bladder irrigation after TURP

5. Acute Kidney Injury

Cause	Examples
Prerenal	Hypotension, dehydration, shock, heart failure, sepsis, burns, liver failure, renal-artery stenosis and nephrotoxic BP/NSAID effects.
Intrarenal	Glomerulonephritis, pyelonephritis, lupus, chemotherapy/contrast, heavy metals and direct renal injury.
Postrenal	Obstruction from stones, clots, BPH, malignancy or neurogenic bladder.

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Phase	Assessment / treatment
Oliguric	<400 mL/day, fluid overload, uremia, hyperkalemia, high BUN/creatinine/phosphate, low calcium/GFR; restrict fluids/electrolytes, diuretics and dialysis as ordered.
Diuretic	Urine 4-5 L/day; dehydration, hypotension, low sodium/potassium; replace IV fluids and monitor closely.
Recovery	GFR rises, BUN/creatinine stabilize, strength and cognition improve; may last months.

PRIORITY Pulmonary edema, severe hyperkalemia, metabolic acidosis, pericarditis and uremic encephalopathy are dialysis-level emergencies.

6. Chronic Kidney Disease

- Progressive irreversible loss with GFR <60 for at least 3 months; common causes include diabetes and hypertension.
- Manifestations: hypertension/edema, CHF/crackles, Kussmaul breathing, uremic fetor, anemia/bleeding, GI upset/metallic taste, pruritus/uremic frost, renal osteodystrophy and neurologic changes.
- Restrict sodium, potassium, fluids, phosphorus and protein as ordered; peritoneal dialysis may require increased protein.
- Epoetin alfa for anemia with hemoglobin monitoring; phosphate binders such as sevelamer are given with food; hyperkalemia therapy may include sodium polystyrene as ordered.

7. Hemodialysis

Access	Nursing priority
Temporary central catheter	Short-term; femoral access has higher infection risk and limited hip flexion.
AV fistula	Preferred permanent access; matures 4-6 weeks. Check bruit and thrill; no BP, IV or venipuncture in arm.
AV graft	Ready sooner but higher clot, aneurysm and infection risk.

- Weigh and check vital signs before/after; hypotension is the most common complication.
- Monitor bleeding because anticoagulation may be used. Hold dialyzable/antihypertensive medications only according to orders.
- Disequilibrium syndrome: headache, nausea, confusion, seizure and hypertension during early dialysis; protect airway, institute seizure precautions and treat as ordered.

8. Peritoneal Dialysis

Phase	Process
Infusion	1-2 L dialysate flows by gravity over about 10-20 minutes.
Dwell	Fluid remains in abdomen from about 30 minutes to 8 hours.
Outflow	Bag below abdomen drains waste fluid.

- Use sterile technique. Cloudy outflow suggests peritonitis: stop/notify and send sample.
- For absent outflow, check kinks and reposition. Decreased outflow may reflect constipation.
- Warm dialysate and slow rate for cramps. Insulin may be added; monitor hyperglycemia.

9. Renal Transplant

- Mild hematuria may occur. Expected urine output is often >100 mL/hr initially; output near 50 mL/hr requires notification. Replace losses as ordered.
- Lifelong immunosuppression and infection precautions are required.

Rejection	Timing / findings / response
Hyperacute	Hours-days: fever, graft tenderness and abrupt low/absent urine from mismatch; graft removal.
Acute	Days-weeks: falling urine, weight gain/edema; increase immunosuppression as ordered.
Chronic	Months-years: rising BUN/creatinine and falling GFR; dialysis or retransplant.

FINAL NCLEX FOCUS Trend urine output, daily weight, potassium, BUN/creatinine and neurologic/respiratory status. Protect dialysis access and treat cloudy peritoneal effluent as infection.